TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	89.0 ± 2.10
Correlation Stability	93.1 ± 0.54
Distribution Consistency	90.4 ± 2.73
Entropy Preservation	79.6 ± 0.49
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	98.8 ± 0.75
Feature Accuracy	93.2 ± 1.99
Novelty Score	0.8 ± 0.01

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Dataset Audit: heart_failure_clinical_records.csv

1. Dataset Portrait

Attribute	Value
Rows	299
Columns	13
Numeric features	13
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	60.834	59.967	-0.867	1.4%	OK
anaemia	0.431	0.335	-0.097	22.4%	Warn
creatinine_phosphokinase	581.839	739.059	+157.219	27.0%	Drift
diabetes	0.418	0.452	+0.034	8.1%	OK
ejection_fraction	38.084	38.724	+0.640	1.7%	OK
high_blood_pressure	0.351	0.326	-0.025	7.1%	OK
platelets	263358.029	265448.138	+2090.109	0.8%	OK
serum_creatinine	1.394	1.313	-0.081	5.8%	OK
serum_sodium	136.625	136.519	-0.107	0.1%	OK
sex	0.649	0.628	-0.021	3.3%	OK
smoking	0.321	0.297	-0.024	7.5%	OK
time	130.261	138.895	+8.635	6.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	89.0 ± 2.10
Correlation Stability (*2)	93.1 ± 0.54
Distribution Consistency (*3)	90.4 ± 2.73
Entropy Preservation (*4)	79.6 ± 0.49
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	98.8 ± 0.75
Feature Accuracy (*10)	93.2 ± 1.99
Novelty Score (*11)	0.8 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8283333333333334	0.7350000000000001	0.7983333333333332	-0.02999999999999998	
F1	0.7517499705786448	0.5412024529278291	0.6914396805701154	-0.06031029000852946	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.835	0.745	0.8083333333333333	-0.02666666666666672	*
F1	0.7547418227770806	0.566990604703608	0.7057148438734797	-0.04902697890360093	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7833333333333333	0.73	0.7666666666666667	-0.01666666666666673	
F1	0.6754715772040935	0.5679961232789141	0.6515999035281675	-0.02387167367592580 8	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7499999999999999	0.7316666666666667	0.7383333333333334	-0.01166666666666681	
F1	0.5667171896628354	0.5003642129709494	0.5244253404540862	-0.04229184920874909	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6333333333333334	0.6333333333333334	0.6333333333333334	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.

ACCURACY	0.825	0.7566666666666666	0.8033333333333333	-0.02166666666666668	
F1	0.7417404771442427	0.5917474246070242	0.6888842868032159	-0.05285619034102688	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8433333333333334	0.7516666666666667	0.8166666666666667	-0.02666666666666666	
F1	0.7603694426488544	0.5526922620254235	0.7077059536526387	-0.05266348899621587	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8066666666666666	0.7583333333333333	0.78	-0.02666666666666665	*
F1	0.6937863401621888	0.54781686178745	0.6407102790498571	-0.05307606111233178	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5116666666666666	0.555	0.4883333333333333	-0.02333333333333338	
F1	0.3515866832806742	0.2017906523414479	0.1937694977489583	-0.1578171855317156	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6133333333333335	0.6083333333333333	0.6183333333333333	0.00499999999999992	
F1	0.007142857142857143	0.12910738246222114	0.09390476190476191	0.08676190476190476	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
2	55.0	0.0	7861.0	0.0	38.0	0.0	263358.03	1.1	136.0	1.0
3	65.0	0.0	146.0	0.0	20.0	0.0	162000.0	1.3	129.0	1.0
4	50.0	1.0	111.0	0.0	20.0	0.0	210000.0	1.9	137.0	1.0
5	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
6	90.0	1.0	47.0	0.0	40.0	1.0	204000.0	2.1	132.0	1.0
7	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0
8	60.0	1.0	315.0	1.0	60.0	0.0	454000.0	1.1	131.0	1.0
9	65.0	0.0	157.0	0.0	65.0	0.0	263358.03	1.5	138.0	0.0
10	80.0	1.0	123.0	0.0	35.0	1.0	388000.0	9.4	133.0	1.0
11	75.0	1.0	81.0	0.0	38.0	1.0	368000.0	4.0	131.0	1.0
12	62.0	0.0	231.0	0.0	25.0	1.0	253000.0	0.9	140.0	1.0
13	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
14	50.0	1.0	168.0	0.0	38.0	1.0	276000.0	1.1	137.0	1.0
15	49.0	1.0	80.0	0.0	30.0	1.0	427000.0	1.0	138.0	0.0
16	82.0	1.0	379.0	0.0	50.0	0.0	47000.0	1.3	136.0	1.0
17	87.0	1.0	149.0	0.0	38.0	0.0	262000.0	0.9	140.0	1.0
18	45.0	0.0	582.0	0.0	14.0	0.0	166000.0	0.8	127.0	1.0
19	70.0	1.0	125.0	0.0	25.0	1.0	237000.0	1.0	140.0	0.0
20	48.0	1.0	582.0	1.0	55.0	0.0	87000.0	1.9	121.0	0.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	61.690948693598	1.0	65.0	1.0	14.0	0.0	292966.0	0.8171033602662485	146.0	0.0
2	49.123243837946966	0.0	597.0	1.0	55.0	0.0	370210.0	1.0332769877574968	136.0	0.0
3	54.490734228465776	1.0	117.0	0.0	36.0	0.0	394704.0	0.8710165536295783	139.0	0.0
4	51.887925477132601	0.0	1500.0	0.0	25.0	1.0	322794.0	2.1132101885445342	138.0	0.0
5	58.809011600411909	1.0	1180.0	0.0	39.0	1.0	370431.0	1.615344031633973	139.0	0.0
6	70.390544902172201	1.0	458.0	1.0	50.0	1.0	234085.0	1.1893086038285009	132.0	0.0
7	45.0	1.0	66.0	1.0	25.0	0.0	233000.0	0.8	135.0	1.0
8	70.0	0.0	161.0	0.0	25.0	0.0	244000.0	1.2	142.0	0.0
9	58.0	1.0	400.0	0.0	40.0	0.0	164000.0	1.0	139.0	0.0
10	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	57.607508699771024	0.0	567.0	0.0	43.0	0.0	310193.0	0.6786235160749732	140.0	0.0
2	70.589468141313409	0.0	143.0	0.0	35.0	1.0	262833.0	1.6791909473397855	140.0	1.0
3	32.766422967337308	1.0	86.0	0.0	44.0	1.0	397987.0	2.3483040432968645	132.0	0.0
4	60.1273694103711	1.0	472.0	0.0	36.0	1.0	183919.0	0.8329563655133414	136.0	0.0
5	38.263522213388608	1.0	259.0	1.0	22.0	0.0	307892.0	1.3511021538149113	129.0	0.0
6	66.701086590386604	0.0	181.0	1.0	41.0	0.0	168255.0	2.4884265089406967	130.0	0.0
7	60.0	0.0	2656.0	1.0	30.0	0.0	305000.0	2.3	137.0	1.0
8	45.0	1.0	130.0	0.0	35.0	0.0	174000.0	0.8	139.0	1.0
9	45.0	0.0	582.0	1.0	55.0	0.0	543000.0	1.0	132.0	0.0
10	65.0	1.0	59.0	1.0	60.0	0.0	172000.0	0.9	137.0	0.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	58.89995505934846	1.0	61.0	1.0	53.0	0.0	285878.0	0.23891946413538212	141.0	1.0
2	56.71442082093333 4	0.0	144.0	0.0	47.0	1.0	200593.0	2.8675271698191693	144.0	1.0
3	82.70478336488985	0.0	28.0	1.0	29.0	0.0	236841.0	0.053079895234174446	137.0	1.0
4	47.01470853435033	0.0	1571.0	1.0	60.0	0.0	339411.0	3.2088202151360172	127.0	1.0
5	66.29186166056252	0.0	-17.0	1.0	30.0	0.0	128403.0	1.9727597444108884	133.0	1.0
6	60.70508922650333 26	1.0	814.0	1.0	23.0	1.0	333704.0	2.397595764331814	126.0	1.0
7	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0
8	79.0	1.0	55.0	0.0	50.0	1.0	172000.0	1.8	133.0	1.0
9	65.0	1.0	160.0	1.0	20.0	0.0	327000.0	2.7	116.0	0.0
10	70.0	0.0	92.0	0.0	60.0	1.0	317000.0	0.8	140.0	0.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	72.00265201053618	0.0	-839.0	0.0	42.0	1.0	204097.0	1.7505289575401215	133.0	0.0
2	66.76878745117358	0.0	265.0	1.0	35.0	1.0	196393.0	1.107223236380157	131.0	0.0
3	67.2918635269383	1.0	357.0	1.0	61.0	1.0	235398.0	0.949072821843415	142.0	1.0
4	48.02148499842732 6	0.0	1695.0	1.0	28.0	1.0	214984.0	1.6295296456331674	122.0	1.0
5	66.60109000611523	1.0	514.0	0.0	49.0	0.0	123967.0	0.7927886721570836	135.0	0.0
6	73.52296333729501	1.0	814.0	1.0	31.0	0.0	227926.0	2.453697339841561	143.0	1.0
7	48.0	1.0	131.0	1.0	30.0	1.0	244000.0	1.6	130.0	0.0
8	95.0	1.0	371.0	0.0	30.0	0.0	461000.0	2.0	132.0	1.0
9	45.0	1.0	981.0	0.0	30.0	0.0	136000.0	1.1	137.0	1.0
10	50.0	0.0	582.0	1.0	38.0	0.0	310000.0	1.9	135.0	1.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	69.6228312768281 4	1.0	2164.0	0.0	31.0	1.0	66848.0	5.77684420202472 5	131.0	1.0
2	41.9153255148528 1	1.0	403.0	0.0	38.0	0.0	219983.0	0.89237118650184 07	136.0	1.0
3	56.7782916730304 3	0.0	262.0	0.0	36.0	0.0	286503.0	1.83130378640014 5	140.0	1.0
4	57.1117396278704 9	1.0	1843.0	0.0	29.0	1.0	309390.0	1.31017113035323 22	137.0	0.0
5	79.3661602141056 5	1.0	222.0	0.0	39.0	1.0	183661.0	1.08374663680210 94	138.0	1.0
6	56.7346806666724 96	0.0	1157.0	0.0	43.0	0.0	53978.0	1.87592499637770 78	137.0	1.0
7	78.0	0.0	224.0	0.0	50.0	0.0	481000.0	1.4	138.0	1.0
8	70.0	0.0	582.0	0.0	20.0	1.0	263358.03	1.83	134.0	1.0
9	61.0	0.0	248.0	0.0	30.0	1.0	267000.0	0.7	136.0	1.0
10	40.0	0.0	478.0	1.0	30.0	0.0	303000.0	0.9	136.0	1.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	55.6908307503131	0.0	2985.0	1.0	29.0	1.0	276691.0	0.45571457339266 33	142.0	1.0
2	64.0568276323389 4	1.0	14.0	0.0	36.0	1.0	354277.0	1.39854085291545 46	141.0	0.0
3	61.5493112725124 4	0.0	-365.0	0.0	37.0	1.0	463894.0	1.16210168068925 32	136.0	1.0
4	60.1588806514800 25	0.0	2593.0	0.0	44.0	0.0	225883.0	1.26842376955582 36	136.0	0.0
5	74.0834049912142 7	0.0	1121.0	1.0	23.0	1.0	209465.0	1.24959588156482 76	134.0	1.0
6	45.4632343013117 15	0.0	830.0	0.0	48.0	0.0	242559.0	0.93688601367437 82	139.0	0.0
7	81.0	0.0	4540.0	0.0	35.0	0.0	231000.0	1.18	137.0	1.0
8	70.0	0.0	2695.0	1.0	40.0	0.0	241000.0	1.0	137.0	1.0
9	70.0	0.0	835.0	0.0	35.0	1.0	305000.0	0.8	133.0	0.0
10	80.0	0.0	776.0	1.0	38.0	1.0	192000.0	1.3	135.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	80.2114217639648 6	0.0	184.0	0.0	42.0	0.0	130528.0	1.08360155308023 68	130.0	1.0
2	86.6538319835037	0.0	2516.0	0.0	42.0	0.0	293048.0	2.38242824599183 3	136.0	1.0
3	47.4550809415502	0.0	633.0	1.0	56.0	0.0	320797.0	2.21179126444675 14	133.0	0.0
4	50.0678344110028	0.0	156.0	1.0	31.0	0.0	182517.0	1.32685991760873 97	136.0	1.0
5	70.9285502926812 4	1.0	5.0	1.0	28.0	1.0	138275.0	1.57733937341505 96	143.0	0.0
6	50.2423930357304 2	0.0	1057.0	0.0	32.0	0.0	422429.0	1.01524338432990 05	141.0	1.0
7	75.0	0.0	582.0	1.0	30.0	1.0	263358.03	1.83	134.0	0.0
8	53.0	0.0	207.0	1.0	40.0	0.0	223000.0	1.2	130.0	0.0
9	60.0	1.0	2281.0	1.0	40.0	0.0	283000.0	1.0	141.0	0.0
10	59.0	1.0	280.0	1.0	25.0	1.0	302000.0	1.0	141.0	0.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	78.6307719717234 5	0.0	144.0	0.0	32.0	1.0	323726.0	0.21432277941699 73	127.0	0.0
2	59.9108868379521 9	0.0	181.0	0.0	41.0	0.0	182164.0	1.81876623808383 96	132.0	0.0
3	53.2657147290088 84	1.0	2362.0	1.0	40.0	0.0	411222.0	0.72053987257976 2	139.0	1.0
4	55.7425450523246	1.0	299.0	1.0	46.0	0.0	198756.0	1.42134145852078 75	142.0	1.0
5	70.9658702193588 6	1.0	-6.0	0.0	11.0	1.0	243751.0	1.56598058043871 84	141.0	1.0
6	75.7168177924844 3	1.0	218.0	0.0	50.0	0.0	383249.0	3.05173751975312 9	134.0	1.0
7	65.0	0.0	224.0	1.0	50.0	0.0	149000.0	1.3	137.0	1.0
8	80.0	0.0	898.0	0.0	25.0	0.0	149000.0	1.1	144.0	1.0
9	66.0	1.0	72.0	0.0	40.0	1.0	242000.0	1.2	134.0	1.0
10	75.0	1.0	582.0	0.0	30.0	0.0	225000.0	1.83	134.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	39.939393146399574	1.0	915.0	0.0	22.0	0.0	511995.0	1.6482400001159307	140.0	0.0
2	65.184968797241821	1.0	244.0	1.0	25.0	0.0	261558.0	2.5746743209821217	129.0	0.0
3	54.221432508467921	0.0	926.0	0.0	32.0	0.0	228772.0	0.8332372862450846	132.0	1.0
4	42.309877236947024	1.0	-48.0	0.0	39.0	1.0	261442.0	0.5588089598750924	139.0	1.0
5	62.091558808462732	1.0	298.0	0.0	53.0	0.0	389585.0	6.31976786734729	135.0	1.0
6	43.191336271869209	0.0	624.0	0.0	30.0	0.0	276703.0	0.9147479181897716	139.0	1.0
7	55.0	0.0	572.0	1.0	35.0	0.0	231000.0	0.8	143.0	0.0
8	60.0	0.0	68.0	0.0	20.0	0.0	119000.0	2.9	127.0	1.0
9	70.0	0.0	97.0	0.0	60.0	1.0	220000.0	0.9	138.0	1.0
10	53.0	1.0	1808.0	0.0	60.0	1.0	249000.0	0.7	138.0	1.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	anaemia	creatinine_phosphoki	diabetes	ejection_fractio	high_blood_pressu	platelets	serum_creatinine	serum_sodium	sex
1	92.022030950596811	1.0	272.0	1.0	60.0	0.0	142097.0	0.7832741656457204	136.0	0.0
2	57.591626213733311	0.0	746.0	1.0	19.0	0.0	611162.0	0.9120760593250713	138.0	1.0
3	56.074627287839609	0.0	356.0	1.0	40.0	0.0	251457.0	1.4168710296797518	129.0	0.0
4	65.9414353706859	1.0	85.0	0.0	23.0	1.0	216387.0	7.1255902179323805	134.0	0.0
5	47.9983294063268	1.0	160.0	1.0	24.0	1.0	305853.0	2.5059564235236764	130.0	0.0
6	51.183155634845505	0.0	251.0	0.0	27.0	0.0	312814.0	2.2358721513110464	143.0	0.0
7	60.0	0.0	897.0	1.0	45.0	0.0	297000.0	1.0	133.0	1.0
8	75.0	0.0	582.0	0.0	20.0	1.0	265000.0	1.9	130.0	1.0
9	45.0	1.0	1876.0	1.0	35.0	0.0	226000.0	0.9	138.0	1.0
10	75.0	1.0	246.0	0.0	15.0	0.0	127000.0	1.2	137.0	1.0